Quorum Intersection and Fenced Cutover for Agent Tools

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Abstract

Two configurations can each satisfy their own quorum rule while permitting disjoint certificates across a membership change. We give exact minimum intersections for direct, staged, and joint threshold certificates. A single median determines the joint minimum. For two stages that may coexist, these minima determine whether the common signers must include someone outside the fault budget. Exhaustive signer-set checks verify the formulas on small committees. A separate finite model tests how a sink can end old authority before accepting a new epoch while preserving the record of earlier irreversible effects. For collective effects, closure and prior absence depend on different families of sink sets. Exact evidence cuts identify when partial acknowledgments suffice. A local SQLite experiment exercises transaction ordering and interrupted acknowledgments.

Summary and scope

Question. How much overlap and terminal evidence are needed when changing the signers that authorize an effect?

Finding. Exact threshold intersections characterize whether certificate pairs must share an honest signer. Sink closure prevents later old effects. Final occupancy evidence is separately needed to avoid duplicating an earlier irreversible effect.

Evidence and scope. The overlap formulas and cutover rule are supported by exact set and interleaving checks. They apply to authorization gates that enforce durable signer and sink rules, without establishing a deployed consensus protocol.

1 What needs to intersect

Let the old committee A have size m and the new committee B have size n, with $s = |A \cap B|$. The old and new thresholds are a and b, respectively, each positive and no greater than its committee size. A joint certificate is a subset of $A \cup B$ with at least a signers in A and at least b in B. A signer in the intersection contributes to both thresholds.

Raft uses majorities in both configurations during joint consensus [1]. Byzantine quorum systems require an intersection containing sufficiently many correct signers [2]. Here the same conflict domain is enforced across configurations, at most f signer identities may equivocate across that domain, and every other signer durably refuses to sign conflicting effects. Under these assumptions, a pair of certificate families rules out conflicting certificates exactly when its minimum intersection exceeds f. If the minimum is at most f, the common signers in a minimizing pair may all equivocate and the disjoint remaining signers can each sign one effect.

2 Exact thresholds

Write $[x]_+ = \max(0, x)$, $p = m - s$, and $q = n - s$. Let I_{AB}, I_{AJ}, I_{JB}, and I_{JJ} denote the minimum signer intersections for old/new, old/joint, joint/new, and joint/joint.

Theorem 1. *The first three minimum intersections are*

$$\begin{aligned} I_{AB} &= [a + b - (m + n - s)]_+, \\ I_{AJ} &= [a + \max(a, b - q) - m]_+, \\ I_{JB} &= [b + \max(b, a - p) - n]_+. \end{aligned}$$

For the joint/joint minimum define $h = \max(0, a - p, b - q)$, $\alpha = 2a - p$, *and* $\beta = 2b - q$. *Let* T_* *be the median of* (s, α, β) *clamped to* $[2h, 2s]$. *Then*

$$I_{JJ} = [T_* - s]_+ + [\alpha - T_*]_+ + [\beta - T_*]_+.$$

Proof. For direct overlap, fill the exclusive parts first. The smallest necessary shared counts are $[a - p]_+$ and $[b - q]_+$. Two subsets of the shared pool with those sizes can overlap by as little as their summed size minus s, floored at zero. This simplifies to the formula for I_{AB}.

A joint certificate must contain at least $\max(a, b - q)$ signers from A. Fill B's exclusive part as far as possible, then meet the remaining B threshold from the shared pool and complete the A threshold. This attains the lower bound. An old certificate can take as many of its signers as possible from the complement of that set in A, which proves the formula for I_{AJ}. Exchanging the committee roles gives I_{JB}.

For two joint certificates with shared counts c_1, c_2, the exclusive counts can be chosen as $[a - c_i]_+$ in the old pool and $[b - c_i]_+$ in the new pool. The shared counts range from h to s. Let $T = c_1 + c_2$. The least overlap in the shared pool is $[T - s]_+$. The sum of exclusive old counts is at least $[2a - T]_+$, and the sum of exclusive new counts is at least $[2b - T]_+$. Both bounds are attained together by taking $c_1 = \lfloor T/2 \rfloor$ and $c_2 = \lceil T/2 \rceil$. For each pool we can choose the two subsets to attain their minimum overlap, independently of the other pools. It follows that the exact minimum for fixed T is $[T - s]_+ + [\alpha - T]_+ + [\beta - T]_+$. The slope starts at -2 and increases by one at each breakpoint s, α, β, counting multiplicities. A median minimizes the expression, and clamping it to the feasible interval preserves optimality. All endpoints and breakpoints are integers, so the minimum is attained at a feasible integer. □

3 Availability and a single equivocator

A joint certificate remains possible under every withholding of at most f signers, with $f < \min(m, n)$, exactly when $a \le m - f$ and $b \le n - f$. Sufficiency follows because enough signers remain in each committee. Necessity follows by placing all f omissions inside one committee. Among thresholds meeting these inequalities, the largest ones maximize the direct minimum, giving $[s - 2f]_+$. Thus concurrent old and new authorization with a guaranteed honest common signer under these availability conditions requires $s \ge 3f + 1$.

The same-epoch minima are $[2a - m]_+$ for two old certificates and $[2b - n]_+$ for two new certificates. Suppose a staged cutover permits old and joint certificates to coexist, then joint and new, and a fence rules out old and new coexistence. With the largest available thresholds $a = m - f$ and $b = n - f$, the old/joint and joint/new minima equal the two same-epoch minima. The joint/joint minimum is at least their maximum. Hence $m, n \ge 3f + 1$ is necessary and sufficient for all allowed pairs to have an honest intersection under the availability requirement, regardless of

shared membership. The necessity comes from each configuration's own pair of certificates. The conclusion relies on enforcing the permitted coexistence schedule at the sinks.

For the old committee $\{A, B, C\}$ and the new committee $\{C, D, E\}$ with majority thresholds, the exact minima in the order above are $(0, 1, 1, 1)$. The two valid joint certificates $\{A, C, D\}$ and $\{B, C, E\}$ intersect only at C. A single equivocator can therefore defeat the honest-intersection requirement even for two joint certificates. For disjoint four-member committees with threshold three, the minima are $(0, 2, 2, 4)$. With one potentially faulty signer, adjacent phases have an honest intersection, but old and new alone still have none.

4 Closing the old epoch at the effect sinks

Chubby's sequencers attach a generation to a lock and allow the effect server to reject stale use [3]. In the following model, the authorization service knows the complete set of sinks that may still commit an old effect. It freezes issuance of old rights or enforces a ceiling on every old right's lifetime before activation. Each trusted sink checks the epoch and expiry at its irreversible commit point. The adversary may delay requests, evidence, and epoch messages.

For sink i, let e_i be a certified hard upper cutoff, so all its old commits occur strictly before e_i. Let b_i be the time at which the coordinator first receives proof of the sink's irreversible old-epoch fence. The fence and commit decisions share a durable order. A missing bound is represented by $+\infty$. Under these evidence rules, the earliest time at which all old authority is known closed is

$$t_{\text{close}} = \max_i \min(e_i, b_i).$$

At that time each sink has either passed its cutoff or supplied proof of a fence. Before that time there is a sink with neither form of evidence. If an otherwise eligible old request can still commit there, delaying the fence and delivering the request yields a history consistent with what the coordinator knows. This makes the bound exact for these observations when no other mechanism can exclude such a request. When the coordinator knows only an interval for the current time, it uses the lower endpoint to establish that expiry has passed.

For example, take two sinks with hard old cutoffs $(9, 13)$ and received fence proofs at times $(11, 8)$. The coordinator has closure evidence for both at time nine. A prior absence report from the first sink may still be stale, since it could have committed between the report and time nine. This remains possible even if the old and new signer sets each met their own quorum rule.

Closing the old epoch prevents future old commits. A new commit that must be unique across both epochs also needs evidence about the past. For each sink, require a terminal record for the conflict key. Such a record reports whether an old effect already committed and is produced atomically with the fence or from an authoritative snapshot after closure. Bind it to the sink, epoch, and conflict key. Only a full set of absent records admits the new effect. An unknown record leaves the key blocked, even after expiry.

To see sufficiency, an absent terminal record rules out a prior commit at its sink, and closure rules out any later old one. The argument applies to every sink that could have committed the key. For necessity in this observation model, a missing terminal record can leave two indistinguishable histories, one with a prior commit and one without it. An absence observation from before closure may be followed by a commit before the fence or expiry. The same test can use an authoritative shared journal if it provides equivalent complete evidence. The rule concerns uniqueness of the specific irreversible effect. It does not undo a committed effect or grant new authority on its own.

4.1 Partial evidence for collective effects

Some tools can commit an effect only when several sinks remain live under the old epoch. Let $\mathcal{H}$ be a nonempty family of nonempty sink sets. Every old commit requires all the members of some $H \in \mathcal{H}$ to be live at its irreversible commit point. Write $c_i = \min(e_i, b_i)$ for the time when closure at sink i is certified. A missing certificate and cutoff give $c_i = +\infty$.

Past effects need a separate assumption. Let $\mathcal{R}$ be the nonempty family of possible durable record footprints for one old commit on the conflict key. Each footprint is nonempty. Write a_i for the time a trustworthy terminal absence record from sink i is available. An unknown or positive record has $a_i = +\infty$. The absence record covers the key and epoch and rules out a durable old commit mark at that sink. It must cover the entire old epoch, including in-flight effects and any delayed replication of the mark.

Theorem 2. *Under these evidence rules, closure and prior absence are certified respectively by times*

$$\max_{H \in \mathcal{H}} \min_{i \in H} c_i, \qquad \max_{R \in \mathcal{R}} \min_{i \in R} a_i.$$

The first bound is exact when each unhit live set can still commit in some history consistent with the observations. The second is exact when a footprint disjoint from all observed terminal absences can still belong to a past commit consistent with those observations.

Proof. After the first bound, the closed sinks meet every required live set. An old commit would require a live sink in that closed set, which is impossible. Before the bound, an unhit live set remains. The stated realizability assumption supplies a continuation with an old commit. For past effects, every possible footprint meets the set of observed terminal absences after the second bound. A prior commit would then contradict an absence record. Before that bound, there is a footprint containing no such record, and the realizability assumption supplies a compatible old commit. The first time every set is hit is the latest of the first-hit times, giving both formulas. □

With n sinks and every q-sink subset allowed as a live set, $n - q + 1$ closure proofs suffice. In a five-sink example, take $q = 3$ and closure times $(2, 5, 8, 11, +\infty)$. Closure is certified at time eight. If every old effect leaves a durable mark on some two-sink subset, and terminal absence times are $(3, 6, 9, 12, +\infty)$, prior absence is certified at time twelve. These tests together permit a unique new effect at time twelve if the new authority and its other gates already hold. A sink can remain unreachable.

Cached signatures cannot count as live participation at commit. For a two-out-of-three gate, a sink can sign at time one and close at time two. Another closes at time three. At time four the last sink can add a valid old signature to the saved one. A gate that checks only the signatures can still accept the old effect. The live-set calculation applies only if the commit protocol itself enforces the required epoch checks at linearization.

5 Reproducible checks

The original three-by-three fixture had five disjoint pairs among nine old/new majority pairs. The generalized check compared the closed forms with enumerated signer sets on 74 committee layouts. All 3,216 minimum-overlap comparisons and 3,468 availability checks agreed. This is a finite check of the certificate arithmetic.

A separate state search modeled two sinks, one conflict key, finite deadlines, and delayed terminal records. The proposed rule reached 2,677 states with neither an old commit after activation nor

a new commit duplicating an old one. Requiring closure from only one sink and allowing a fence to roll back each admitted an old effect after activation. Taking expiry as proof of prior absence and using pre-closure absence each admitted a duplicate. The receipt contains a shortest witness for each change to the model. The search does not cover network liveness or the correctness of a physical effect sink.

The collective-effect check compared both time bounds with a stepwise oracle on 512 cases. A separate enumeration of possible record footprints checked 6,336 absence patterns. It certified 2,899 patterns with fewer than all sinks reporting. A cached-vote control passed the signature threshold after live closure, as predicted. These counts describe finite models of the declared commit and record rules.

5.1 A local transaction and crash experiment

We implemented the effect boundary as a database row with a conflict key. An effect starts a SQLite write transaction, checks the epoch and key, and inserts the row before committing. Fencing updates the accepted epoch and reads the key in another write transaction. The coordinator receives the terminal record only after that transaction commits. In the test, the committed row is the complete effect.

We ran the same cases with SQLite 3.54.0 in rollback-journal and write-ahead-log modes, with full synchronous writes. An old transaction that acquired the write lock first prevented the fence from acquiring it. After the old commit, the fence observed the occupied key. Once the fence committed, the old epoch could no longer insert a different key. A fresh authorized key could still commit under the new epoch.

Eight child processes exited at specified points before or after an effect or fence commit. Within each pair, the caller saw the same response and exit code. Reopening the store distinguished the outcomes. A pre-commit effect exit left the key absent. A post-commit exit left it occupied. Likewise, reopening after a post-commit fence exit preserved the new epoch and rejected an old request. Reopening after a pre-commit fence exit retained the old epoch until a later successful fence.

Two negative controls exercised the modeled failures. A process that cached authority before fencing could insert an old-epoch row after the fence if its later transaction skipped the epoch check. A coordinator that inferred absence from a lost acknowledgment could commit the same key in a separate store, whose local key index could not see the original row. The correct route admitted a new write only after a terminal absence record from the old store. These tests cover process exit and local database state. They leave authentication, hardware power loss and effects outside the database to separate experiments.

6 Application and next test

An agent tool that authorizes sensitive writes can compare the minimum overlap of every certificate pair its sink may accept. If the tool uses staged authorization with small shared membership, it needs a durable fence that prevents old-only certificates from authorizing effects once new-only authority is active. Replayed certificates, disconnected sinks, and signer-state loss can invalidate that ordering. The arithmetic assumes the same durable conflict rule at every phase and says nothing about whether such a fence, authenticated key change, or network progress guarantee has been implemented.

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References

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